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Prediction of human phosphorylated proteins by extracting multi-perspective discriminative features from the evolutionary profile and physicochemical properties through LFDA

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ABSTRACT

Protein phosphorylation is an emerging post-translational modification, which critically involved in the intracellular process of the human body by controlling diverse functions ranging from cell growth to metabolism. The existing experimental methods for identifying phosphorylated proteins are overpriced and resource-intensive; thus, it is necessary to develop a fast and accurate computational method to address the problem. Here we report a novel predictor HPhosPPred, a phosphorylated protein prediction method that is incorporating highly discriminative evolutionary and physicochemical information conserved in protein primary motifs, namely pseudo-position specific scoring matrix, the auto-covariance transformation of the position-specific scoring matrix and normalized moreau-broto auto-correlation. Further, to boost up the generalization capability of HPhosPPred, we used local fisher discriminant analysis as a dominant feature selection strategy for eliminating redundant and noise patterns from the extracted features. Finally, the optimized features feed to support vector machine with radial basis function kernel to predict phosphorylated proteins. As evident from the results, the proposed method achieved promising performance with an accuracy of 80.68%, sensitivity of 84.63%, specificity of 73.67%, and Matthew's correlation coefficient of 0.581 using rigorous leave-one-out-cross-validation test and 10-fold cross-validation test. The empirical outcomes demonstrate that the developed model outperformed the existing state-of-the-art methods. Furthermore, our analysis reveals that the proposed tool can help detect unseen phosphorylated proteins in particular and proteomics research in general. The source code and dataset are publicly available at https://github.com/saeed344/HPhosPPred.

1. Introduction

Post-translational modifications (PTMs) refer to the covalent addition and enzymatic modification of protein during or after protein biosynthesis, which performs crucial roles in regulating protein functions and modifying gene expression [1,2]. PTMs are responsible for regulating cell signaling (sensing and transduction) and expansion of genetic code information [3,4]. It can determine the changes in protein attributes by adding a modified group or proteolytic cleavage to amino acids [5]. Currently, the abundance of laboratory-based verified samples of PTMs has been detected. From broad spectrum of PTMs, protein phosphorylation primarily on serine (Ser), threonine (Thr), or tyrosine (Tyr) sites is of considerable significance with various biological activities such as signal transduction, DNA repair, metabolism, survival differentiation, translation and transcription [6–8]. Human phosphorylation proteins play a fundamental part in controlling protein conformation, and cellular functions [9], including cell growth, survival differentiation, metabolism, and apoptosis, and its deregulation is responsible for different diseases [10]. In recent years, several studies describing experimentally validated modifications on Ser/Thr/Tyr sites, like sulfation and nitration, O-linked glycosylation and acetylation (O-glcNAc, O-GalNAc) [11–13]. Keep in mind the potential of PTM sites in various diseases [14], and to understand the underlying molecular mechanisms for designing new drugs, a human protein sequence can be phosphorylated or not is a problem of significant interest. Therefore, an effective and convenient method is required for the analysis of protein phosphorylation. The

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mechanism of phosphorylation protein identification usually depends on the protein kinases method [15]. Most common phosphorylated amino acid types, i.e., tyrosine, threonine, and serine, accompany those protein kinases [16,17]. Various approaches have been applied for the protein phosphorylation prediction by quantifying the addition of a negative charge on the protein, which include mass spectroscopy, the sensitive and selective electrochemical, optical detection methodologies [18], phosphor-specific antibody [19], radioisotope labeling [20], dual-mode field-effect devices and nano-plasmonic sensors [21] etc. However, these traditional approaches (wet-lab based) for identification of phosphorylated proteins are time-consuming and labor-intensive. It is worthwhile to mention here that some of the early-mentioned methods require chemical reagents and phosphor-specific antibodies during the experimental process [10,21].

In the post-genomic age, due to the massive growth of protein sequences, determining the phosphorylation possibility of an uncharacteristic protein is critical work, which can lead researchers to find target proteins and disclose their experimental profiles. Computational studies of PTMs are getting more consideration and popularity [22], and numerous approaches have been proposed regarding prediction of PTMs, for example, the identification of protein phosphorylation sites in viruses [23], general phosphorylation site prediction [24]. A multi-label classifier for human phosphorylated proteins [10,25], prediction of phosphorylation sites in human proteins [26], glycosylation, phosphorylation [11], ubiquitination sites [27] and recognizing phosphorylation sites in proteins by combining different pseudo components into an ensemble classifier [28]. Recently, for the prediction of phosphorylated, one o the pioneer predictor named iPhos-PseEvo was developed by Qiu et al. [6]. The iPhos-PseEvo predictor was trained by ensemble random forest with combining the protein sequence evolutionary information into the general pseudo amino acid composition (PseAAC) through the grey-system theory. The proposed, iPhos-PseEvo attained the 71.79% accuracy, 70.06% specificity, and 71.16% sensitivity via leave-one-out-cross-validation (LOOCV) test. Cui et al. [29], recently designed an intelligent sequence-based bioinformatic tool called SK-SOM(Supervised kernel self-organizing maps) for identifying human phosphorylated proteins. The proposed automated method train the model by ensembling multiple SK-SOMwith sequencial features to enhance the prediction performance of the system. Although, much progress has been made for acurate prediction of human phosphorylated proteins, however still room exist for significant improvements in term of feature-engineering, proper feature-slection and intelligent classification algorithm. First, the predictive capability of previous approaches is not enough due to poor protein motifs representation. Secondly, a robust feature optimization algorithm is needed to optimized and select best discrimination feature subset for the final prediction model. Thirdly, selectin of kernel function in classification algorithm has an indispensable influence on the prediction results as each kernel function utilized the distinct procedure for the transformation of data.

In view of the issues mentioned above, we intend to develop an intelligent sequential predictor HPhosPPred, targeting human phosphorylated proteins. In this research, we designed three nominal features extraction descriptors; evolutionary profile based features, i.e., pseudo-position-specific scoring matrix (PsePSSM), the auto-covariance transformation of the position-specific scoring matrix (PSSM-ACT) and physicochemical properties based feature normalized moreau-broto autocorrelation (NMBAC) from phosphorylated proteins sequences. Further, we serially integrated these descriptors and implemented local fisher discriminant analysis (LFDA) as an optimal feature selection method to choose discriminative features. Finally, the optimized feature set is feed into a support vector machine (SVM) with RBF kernel for training the model. The best performance metrics of the model are analyzed under different λ, *lg*, and *lag* values using a 10-fold cross-validation test. We utilized both LOOCV and 10-fold cross-validation tests to perform the experimental analysis. Although previous methods only utilized LOOCV test, in the comparison section, we compared the results of proposed HPhosPPred predictor with state-of-the-art approaches through stringent LOOCV test and also demonstrated the results of the produced predictor via 10-fold cross-validation tests. The comparative results illustrated that the proposed predictor has achieved better prediction performance. Fig. 1 demonstrates the workflow of our developed algorithm HPhosPPred.

The rest of the paper is organized as follows: Section 2 explains materials and methods in detail; section 3 describes the performance evaluation criteria with cross-validation techniques, section 4 presented experimental results, and section 5 concludes the paper with future directions of the research.

2. Material and methods

2.1. Benchmark dataset

To achieve reliable prediction outcome, a standard benchmark dataset is a prime consideration to train and test the computational predictor [30]. Therefore, in this study, we have utilized the dataset from previous work, iPhos-PseEvo [6], for fair and useful comparison. The dataset contains 1770 human phosphorylated proteins, including only serine, threonine, or tyrosine. Qiu et al. originally constructed the dataset., from the UniProtKB/Swiss-Prot database at http://www.ebi.ac.uk/uniprot/released in September 2015, more details are described in their work [6]. Some strict constraints were followed for the construction of this dataset such as (i) experimental assertion was used for evidence (ii) sequences of only human protein was collected (iii) the keyword 'phosphoserine,' 'phosphothreonine,' or 'phosphotyrosine' was used in the advanced search option. After the collection of human protein sequences, some more additional constraints were applied, such as the sequences with amino acid residues length less than 50 and more than 5000. The well-known CD-HIT software [31] is used to reduce the homology bias and redundancy with $\geq$50% for any two phosphorylation protein samples. After these procedures, 1770 human protein sequences were obtained, in which 1132 sequences belong to phosphorylated proteins, and 638 belong to non-phosphorylated proteins. In the phosphorylated proteins sample, 845 are phosphorylated at Ser, 386 at Th, and 249 at Tyr. The benchmark dataset $\mathbb{S}$ can be formulated as;

$$\mathbb{S} = S^{+} \cup S^{-} \tag{1}$$

where S^{+} subset includes the positive sequences of phosphorylated protein and S^{-} subset contains the negative samples of non-phosphorylated protein sequences and $\cup$ represents the symbol for "union" in the set theory.

2.2. Feature extraction

To develop a computation model, one of the challenging tasks is how to formulate amino acid sequences by the appropriate feature representation method that can truly reflect and extract valuable hidden discriminative patterns inside the data [32,33]. In this study, three feature abstraction algorithms are employed to extract discriminative features from primary human phosphorylated proteins, the informative features discovered from evolutionary information via PsePSSM and PSSM-ACT, and physiochemical properties by NMBAC. Details of these techniques are explained in the following sections.

2.3. Evolutionary information features

Position-specific scoring matrix (PSSM) is an evolutionary information method, which is mostly used in motifs representation in biological sequences. In the last few decades, many researchers showed that evolutionary information extracted from the PSSM profile has proved to be essential features for resolving numerous bioinformatics problems, such as in Ref. [33–36]. PSSM profile is generated for every protein

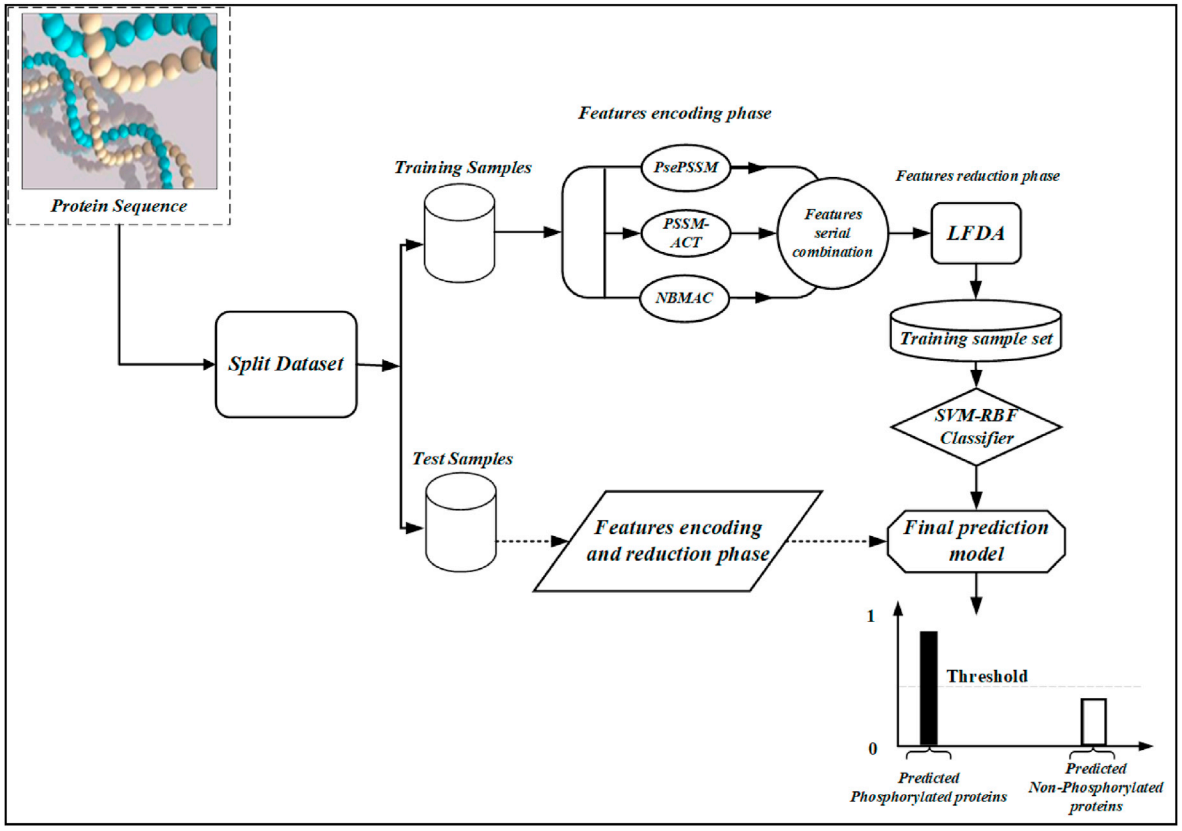


Fig. 1. Framework of the proposed method.

sequence by searching and aligning it to the defective tool called PSI-BLAST for investigation against the Swiss-Prot database [37,38] with parameters $E = 0.001$ and $j = 3$. This PSSM matrix is a log-ad of size $L \times 20$, where L denotes the dimension of the protein (query) amino acid sequence, and 20 is due to the twenty amino acids. The protein sample P can be formulated as;

$$P_{PSSM} = \begin{bmatrix} h_{1,1} & h_{1,2} & \cdots & h_{1,20} \\ \vdots & \vdots & \vdots & \vdots \\ h_{i,1} & h_{i,2} & \cdots & h_{i,20} \\ \vdots & \vdots & \vdots & \vdots \\ h_{L,1} & h_{L,2} & \cdots & h_{L,20} \end{bmatrix} \tag{2}$$

where $h_{i,j}$ indicates the score of the amino acid residue in ith position with the protein sequence being changed to amino acid type j during evolution computing process. After calculating the PSSM matrix, we then normalize each element of PSSM matrix further by standard conversion as follows;

$$h_{i,j} = \frac{h_{i,j}^0 - \overline{h_i^0}}{std\left(\overline{h_j^0}\right)} (i = 1, 2, \ldots, L; j = 1, 2, \ldots, 20) \tag{3}$$

where $h_{i,j}^0$ is the score of evolutionary information directly obtained by PSI-BLAST, $\overline{h_i^0}$ is the mean of $h_{i,j}^0$ over 20 amino acids and standard deviation of $h_{i,j}^0$ is $std(\overline{h_i^0})$. In the benchmark dataset, the length protein sequence "L" is different, and it is not possible to develop such a predictor, which can handle the unusual length of protein sequences. Due to this issue, the protein sequence is converted into a fixed-length feature using the following formula;

$$\bar{P}_{PSSM} = [\overline{h_1 h_2}, \ldots., \overline{h_{20}}]^T \tag{4}$$

$$\overline{h}_j = \frac{1}{L} \sum_{i=1}^{L} h_{i \to j} (j = 1, 2, \ldots, 20) \tag{5}$$

where h_j represent the aggregate score of the amino acid residues in the protein sequence h being substituted for amino acid type j during the biological evolutionary processes [39]. PSSM is an effective way to express protein sequences, but the problem is that different proteins sequence had different lengths when the PSSM was used to represent the evolutionary information of a protein sequence, so all the sequence order information would be lost [40]. In order to partially reflect the local sequence-order effect and to overcome the defect of losing sequence-order information, we used the concept of the pseudo-position-specific scoring matrix (PsePSSM) and auto-covariance transformation (ACT). More details of these methods are provided below:

2.3.1. Pseudo position-specific scoring matrix

Pseudo position-specific scoring matrix (PsePSSM) is the extended version of simple PSSM to capture the additional relevant information from proteins sequence [41]. It has been widely used for the identification of protein-protein interaction [25], anticancer peptide identification [39], prediction of membrane protein [42], protein structure class prediction [43] and predicting protein sub-mitochondrial locations [44]. To keep the local order information of sequence, the proteins sequence h can be represented by PsePSSM as;

$$P^{\lambda}{}_{PesPSSM} = \Big[\tag{6}$$

where

$$Q_j^{\lambda} = \frac{1}{L - \lambda} \sum_{j=1}^{L-\lambda} \left[h_{i \to j} - h_{(i+\lambda) \to j} \right]^2 \left(n = 1, 2, 3, \ldots, 20; \lambda < L \right) \tag{7}$$

where Q_j^1 is the first correlation factor by coupling the most contiguous PSSM scores along the protein chain for the amino acid type j; Q_j^2 that by coupling the second-most contiguous PSSM scores; and so on. When $\lambda = 0$, Q_j^{λ} becomes a zero element and Eq. (5) is degenerated to Eq. (6), which means PsePSSM preserves not only PSSM features but also additional correlation information. Resultantly, $20 + 20 \times \lambda$ dimension vector is produced. Here, in this study, as there is no theoretical justification for the values of λ, thus, after a grid-search approach and empirical evaluation, the value of $\lambda = 9$ is selected, and consequently, a $20 + 20 \times 9 =$ 200-D feature vector is generated.

2.3.2. Auto-covariance transformation of a position-specific scoring matrix

To recuperate the deficiency of the PSSM, where sequence-order information is lost during the evolution process of the protein, we perform autocovariance transformation (ACT) on PSSM, named as PSSM-ACT, to extract partial sequence-order information [45]. In the PSSM-ACT feature encoding approach, the ACT is applied to every PSSM matrix columns to capture the local discriminatory information [46]. It is a valid statistical method used to evaluate sequences of vectors and has been mostly used in the field of bioinformatics for various problems such as prediction of Bioluminescent [47], Bacterial type IV secreted effectors [48] and structural class prediction [49]. ACT variables are capable of escaping producing too many variants [50]. This study employed ACT to transform the variable-length PSSM matrix into a constant vector length. In a given protein sequence, between residues ACT variable draw average interaction with a sequence of *lg* value, where *lg* is the interspace amongst one residue and its *P* protein sequence neighbors. For the ACT, the PSSM matrix of a protein sequence can be calculated as;

$$PSSM - ACT_{(i,lg)} = \frac{1}{L - \lg} \sum_{j=1}^{L-lg} \Big(\tag{8}$$

where

$$\bar{P}_i = \frac{1}{L} \sum_{j=1}^{L} (P_{i,j}) \tag{9}$$

Here $P_{i,j}$ denotes the PSSM matrix, *i*the position of residue, *j* descriptor, and *L* is the size of the protein sequence. The ACT variables number can be calculated as $20 \times lg$, (where $lg = 1, 2, \ldots, 10$), where *lg* is a positive digit that represents the clustered number of transformation and 20 PSSM matrix columns. The PSSM-ACT can provide more information than an individual feature by combining both sequence-order information (ACT features) and the evolutionary information (PSSM composition) of a protein. However, how to set the best value of *lg*, there is no theoretical justification, we randomly tested different values for *lg* using a grid-search method and finally found the best result on $lg = 3$, and thus the final dimensionality of PSSM-ACT feature vector is $20 \times 3 =$ 60-D.

According to the recent study [51], the PSSM information also has shortcomings. The generation of PSSM of a protein depends mainly on the searching dataset. If no homologous sequence is found in the searching dataset, the PSSM will not give exact description, thus leading to wrong prediction. With primary sequence information, our model can obtain such high accuracy, suggesting that the proposed model is more concise and efficient.

2.3.3. Normalized moreau-broto autocorrelation

The protein sequence is a collection of 20 natural amino acid residues that poses different physicochemical properties [52]. These physicochemical properties play a fundamental role in protein function prediction by considering the interaction among amino acids [53–55]. In this study, we employed Normalized Moreau-Broto Autocorrelation (NMBAC) to the physicochemical properties to effectively encode protein sequence using six physicochemical properties [56,57]. These physicochemical properties are net charge index of side chains (NCISC), hydrophobicity (H), solvent-accessible surface area (SASA), polarizability (P2), volumes of side chains of amino acids (VSC) and polarity (P1) of amino acid. The numerical values of these six physicochemical properties against the 20 amino acids are shown in (supplementary file Table S1). These values are normalized as, to zero mean and unit standard deviation by using the follows formula;

$$N_{nm}^{'} = \frac{N_{m,n} - N_m}{S_m} \ (m = 1, 2, 3, \ldots\ldots\ldots, 20; n = 1, 2, \ldots.6) \tag{10}$$

Here $N_{mn}^{'}$ represent *m* property values for amino acid type *n*, N_m denotes the mean value of above 20 amino acids of descriptor value *j* and S_m is the standard deviation values, respectively. A protein sequence can be replaced by vectors, which consist of normalized physicochemical properties values extract by NMBAC. The NMBAC can be calculated by putting these vectors into the following formula;

$$NMBAC_{lag,n} = \frac{1}{L - lag} \sum_{m=1}^{L-lag} \left(p_{m,n} \times p_{m+lag,n}\right) \ \text{where} \ (i = 1, 2, \ldots., L - lag; \ j = 1, 2, \ldots 6) \tag{11}$$

Here *n* represents one physiochemical property of six properties, in *m*th position of protein sequence *P*, *L* is the size of the protein sequence, and the correlation distance between one residue and another is indicating by *lag*. We set the value of *lag* from 1 to 50 as calculated by Guo et al. in Ref. [57] and for each protein sequence, the features vector is $50 \times 6 = 300$. We also estimate the frequency of 20 amino acids, and finally, a protein sequence is represented as a $15 \times 6 + 20 = 110$-D feature vector.

2.4. Features optimization via local fisher discriminant analysis

Feature selection plays an essential role in eliminating redundancy, irrelevant feature and control the overfitting issue of the prediction system [58]. In the present study, we carried out a powerful feature selection technique called local fisher discriminant analysis (LFDA) to select the optimum feature subset. The feature vector for every protein sequence is collected from PsePSSM, PSSM-AC, and NMBAC. These captured features may contain some noise and redundant information with large dimensions. In order to remove noise and redundant information, the LFDA feature selection technique has been employed in the current research work. We also utilized other feature selection methods like AKPCA, Laplacian, and PCA along with LFDA techniques. Here, we only focus on the details of LFDA technique as the comparison suggested that LFDA outperformed other techniques.

LFDA has supervised feature selection method which can calculate the levels of the within-class disperse, and the between-class disperse locally [59]. To explain the working of LFDA, let consider the data matrix of protein sequence be $X = [\chi_1, \chi_2, \ldots, \chi_n], \chi_i \in R^d$, where *n* and *d* denoted the proteins number of samples and extracted feature dimension, respectively. The local inter-class scatter matrix $S^{(b),}$ and intra-class scatter matrix $S^{(w)}$ is calculated using Eqs. (12) and (13) [60];

$$S^{(w)} = \frac{1}{2} \sum_{j=1}^{n} W_{i,j}^{(w)} (\chi_i - \chi_j)(\chi_i - \chi_j)^T \tag{12}$$

$$S^{(b)} = \frac{1}{2} \sum_{j=1}^{n} W_{i,j}^{(b)} (\chi_i - \chi_j)(\chi_i - \chi_j)^T \tag{13}$$

where

$$W_{i,j}^{(w)} = \begin{cases} \frac{A_{i,j}}{n_\ell} & if \quad y_i = y_j = \ell \\ 0 & if \quad y_i \neq y_j \end{cases} \tag{14}$$

$$W_{i,j}^{(b)} = \begin{cases} A_{i,j}\left(\left(\frac{1}{n}\right) - \left(\frac{1}{n_\ell}\right) & if \quad y_i = y_j = \ell \\ \frac{1}{n} & if \quad y_i \neq y_j \end{cases} \tag{15}$$

Here n_ℓ denote the labeled samples in class $y_j \in \{1, 2, 3.....c\}$. The affinity value matrix is denoted by A, $A_{i,j} \in A$ between χ_i and χ_j. Here, we used the affinity value based on the local scaling heuristic [61] which calculated as;

$$A_{i,j} = \exp\left(\frac{\left\|\chi_i - \chi_j\right\|^2}{\sigma_i \sigma_j}\right) \tag{16}$$

Here the parameter $\sigma_i = \left\|\chi_i - \chi_i^{(k)}\right\|$ denotes the local scaling of the adjacent χ_i data sample and $\chi_i^{(k)}$ is the K-nearest neighbor of χ_i. From the literature [59], we know that when K = 7 it can give best results for high-dimensional data; therefore, we select in this study K = 7. The transformation matrix of LFDA can be defined as;

$$T_{LFDA} = \underset{T \in R^{d \times r}}{\operatorname{argmax}} \left[tr\left(\left(T^T S^{(w)}\right)^{-1} T^T S^{(b)} T \right) \right] \tag{17}$$

whereafter calculating the transformation T_{LFDA} the original data matrix converts to reduced data matrix as;

$$Z = T_{LFDA}^{'} X \tag{18}$$

From Eq. (18), Z is the reduced dimensionality matrix with redundant information removed.

2.5. Selection of support vector machine as a classifier

Classification is a supervised machine-learning (ML) technique, where the label of the target class is known during training [36,62]. In this study, we employed five ML-algorithms, namely: support vector machine (SVM), decision tree (DT), random forest (RF), linear discriminative analysis (LDA), and k-nearest neighbor (KNN). After a comparative analysis of these classifiers, we found that with radial basis function (RBF) kernel-based SVM classifier provided noticeable better results. Thus, the final model trained and tested with SVM-RBF due to its great learning and generalization ability. SVM is a robust ML-algorithm for binary classification. It has been broadly used in the field of bioinformatics, data mining, machine learning, and proteomics research problems [63–66]. SVM in binary class problem uses the maximum margin and attempts to separate two classes. It has four kernel functions types, including Sigmoid, Linear, Polynomial, and RBF, which transfer the feature space in high dimension and facilitates the nonlinear classification [67,68]. In this paper, we have compared the performance of four kernels function see Table 06 and choose RBF as kernel of SVM to perform final prediction because of better performance and most widely used [69] which is calculated as;

$$\kappa\left(x_i, y_j\right) = \exp\left(-\gamma \left\|x_i - y_j\right\|^2\right) \tag{19}$$

Here x_i and y_j are the feature vectors of the ith and jth protein sequences, respectively $\gamma = \left(\frac{1}{2\sigma^2}\right)$ is a parameter and σ the kernel width parameter. To receive the optimal results, the regularization parameter C and the kernel parameter γ are optimize using the grid searching strategy via using the LIBSVM software [60], which can freely download from this http://www.csie.ntu.edu.tw/%7Ecjlin/libsvm/.

3. Performance evaluation criteria and cross-validation

In a statistical, computational model, there are three types of cross-validation tests; K-fold cross-validation test, jackknife test (also known as a leave-one-out-cross-validation test (LOOCV)) and independent dataset test [70]. However, the LOOCV test is considered the least arbitrary and most objective one and hence has been broadly accepted and progressively adopted, among the three test methods [40,71]. In the LOOCV test, every sample in the training dataset is treated as an independent test sample, and all the rule-parameters are calculated without counting the one being identified [32]. Although, the LOOCV test gave a unique result for a given benchmark dataset and considered the most effective validation method. However, the computational complexity of the LOOCV test is proportional to the total number of samples in the dataset, thus making it a time-consuming algorithm to implement [72]. In this research work, we have adopted both LOOCV and 10-fold cross-validation test for the performance evaluation of the proposed technique. The authors of the existing predictors, have employed LOOCV test during their research work. Despite the time-complexity of LOOCV test, we have also applied LOOCV test in our research work in order to provide a fair and better comparison with existing approach.

For performance evaluation, we adopted five commonly used indexes in literature: Sensitivity *(SEN)*, Specificity *(SPE)*, Mathew's correlation coefficient *(MCC)*, Accuracy *(ACC)*, and Area under the curve (*AUC*) [72]. These metrics were calculated by the follows formulas;

$$SPE = \frac{TN}{TN + FP}$$

$$SEN = \frac{TP}{TP + FN}$$

$$ACC = \frac{TP + TN}{TP + FP + TN + FN}$$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{20–23}$$

where *TP* (true positive) is correctly predicted, true phosphorylated protein, *TN* (true negative) is correctly predicted, true non-phosphorylated protein, *FP* (false positive) is incorrectly predicted non-phosphorylated protein as phosphorylated protein, and *FN* (false negative) is incorrectly predicted phosphorylated protein as non-phosphorylated protein. In these evaluation indexes, the *SPE* is a rate of all correctly predicted non-phosphorylated among all negative predictions, and the *SEN* (also known as true positive rate) is a rate of correctly predicted phosphorylated protein among all positive samples. *ACC* is the ratio between correct predictions vs. incorrect predictions, which in range 0–100. *MCC* measures the ratio between the observations and prediction. The value of *MCC* ranges from −1 to 1, was near to 1 show an accurate estimate, near to 0 mean that no better than random prediction and −1 means total divergence between observations and predictions. Moreover, for evaluating the generalization performance of the prediction model, the receiver operating characteristic (ROC) curves were also plotted in Fig. 3. The ROC curve is calculated by considering how the trade-off between true positive rate (TPR) and false positive rate (FPR) deviations at a range of different cut-offs of class prediction probability returned by a given classifier.

4. Results and discussion

4.1. The selection of optimal model parameters

In order to develop a high-throughput and effective computational model, optimal parameter selection is always essential in the feature extraction stage. In the current research, we obtained features information from the human phosphorylated proteins sequences using PsePSSM, PSSM-ACT, and NMBAC feature abstraction techniques. It is worthy of mentioning that these descriptors capture the nominal attributes by adjusting the corresponding metrics (λ, *lg*, and *lag*). In case of PsePSSM

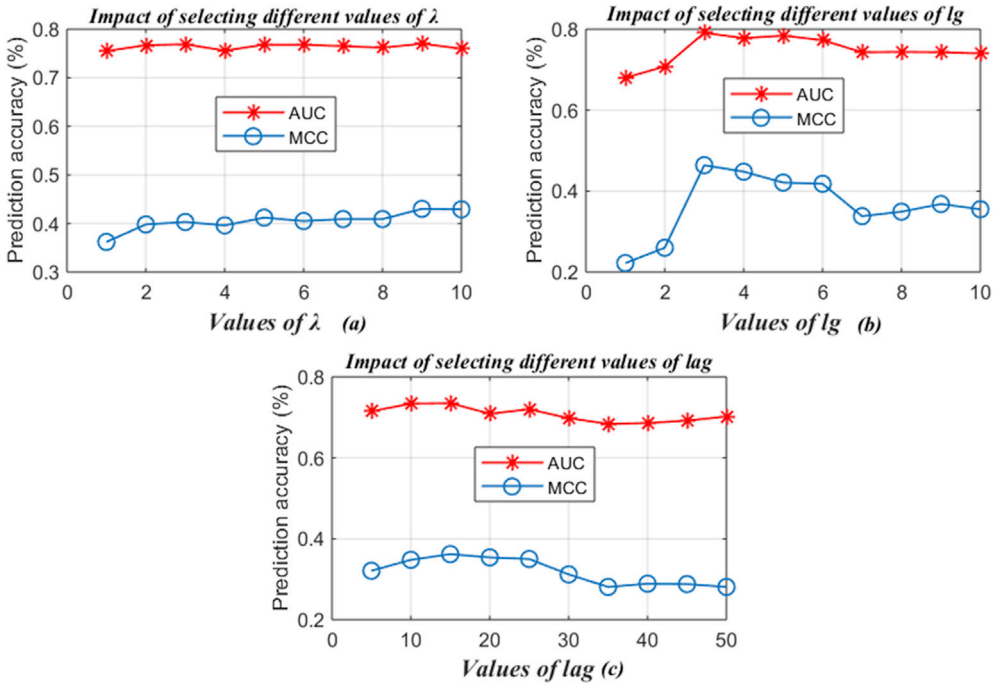


Fig. 2. (a) shows the impact of λ (b) indicates the effects of *lg* and (c) shows the impact of *lag*.

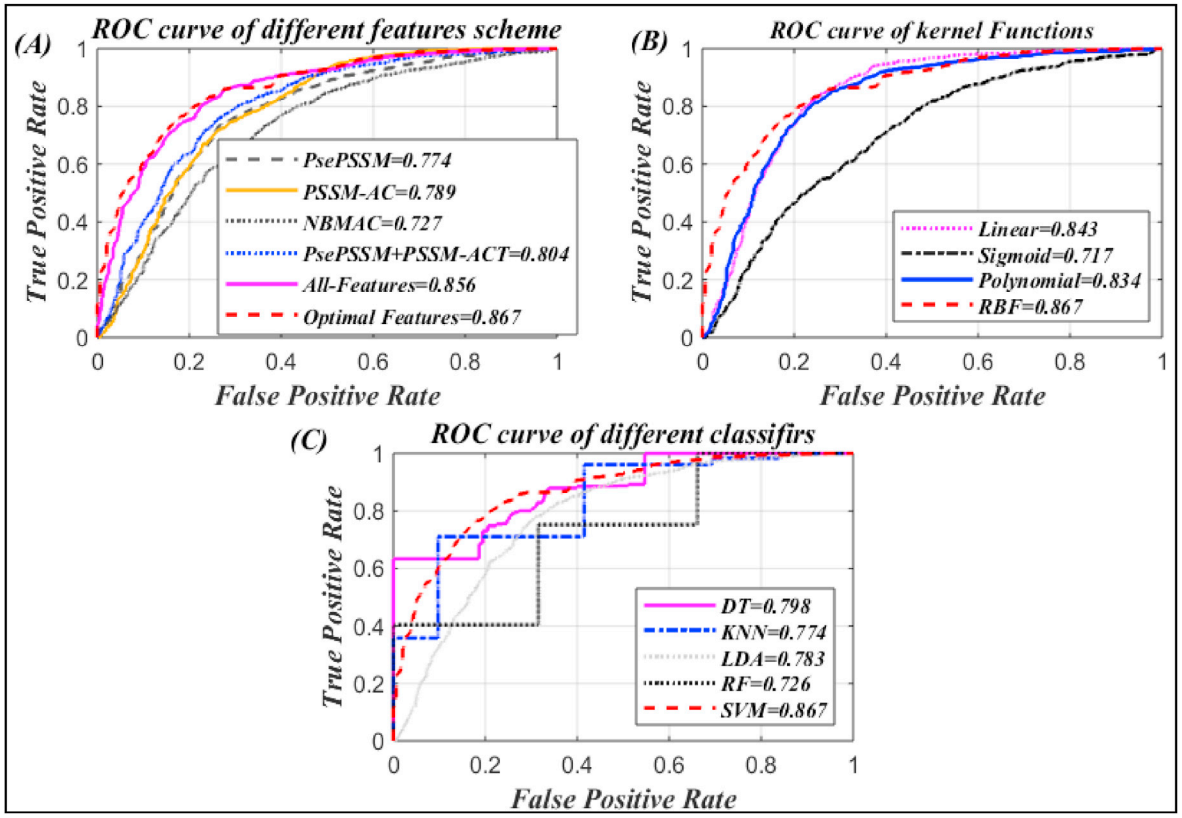


Fig. 3. (A) represents the ROC curve of features scheme, (B) represents the ROC curve of kernel function and (C) represents the ROC curve of five classifier.

and PSSM-ACT algorithms, parameters λ and *lg*, shows the amino acid residues sequence-order information in protein sequence, respectively, while the NMBAC features abstraction parameter *lag* value expressed the distance between two amino acid residues of the protein sequence. The choice of λ, *lg*, and *lag* values have a significant role when the values of λ, *lg*, and *lag* are set too small, extracted feature information will be very less and might lose some essential features information. If the λ, *lg*, and *lag* values are set too high, the feature vector dimension will be too high, which will capture irrelevant information and will affect the performances of model [48,59,73]. However, there is no theoretical

justification exists to adjust specific optimal values for these parameters. The grid-search strategy employed to find optimal values for these parameters and set the values for λ, and *lg* 0 to 10 with a step size of 1 and set the values of *lag* with a step size of 5, from 5 to 50 respectively.

To measure the prediction performance of these feature extraction methods under different values of the corresponding parameter, *SEN, SPE, MCC, ACC,* and *AUC* are used. Table 1, Table 2, and Table 3 are listed the *SEN, SPE, MCC, ACC,* and *AUC* values achieved by different parameters (λ, *lg,* and *lag* values) of PsePSSM, PSSM-ACT, and NMBAC feature scheme respectively. The variations in the *AUC and MCC* values resultant to different *λ, lg, and lag* values in the PsePSSM, PSMM-AC, and NMBAC feature scheme are showed in Fig. 2(a), (b) and (c) respectively.

Table 1 listed the various evaluation metrics using a PsePSSM method to predict human phosphorylated proteins at a different value of λ; the variation was occurring in evaluation indicators when varied the values of λ. We can see from Table 1 that, when the λ is equal to 9, the model has gained the maximum value of *MCC, ACC,* and *AUC* is 74.69%, 0.430 and 0.770 which was 4.09%–0.14%, 18.78%–1.17% and 1.98% to 0.13% respectively higher than achieved by other parameters. To select the optimum parameters of the PsePSSM method, Fig. 2(a) plotted the variation of the *AUC* and *MCC* values when different values of λ were selected. From Fig. 2(a), we observed that the maximum values of *AUC* and *MCC* obtained when the λ value is set nine.

Considering the metrics values in Table 2, using the PSSM-AC algorithm to predict the phosphorylated proteins, when changing the *lg* values, it has a significant impact on the outcomes. The *MCC* and *AUC* values are varying for tuning or changing the values of *lg,* as can be seen from Fig. 2(b). It is illustrated from Fig. 2(b) that the costs of *AUC* and *MCC* initially going up to a maximum peak and then slightly go down as the value of *lg* increased. Between these, when the *lg* is 3, the maximum value of *MCC* and *AUC* achieved 0.791 and 0.464, respectively, which is 16.32%–0.89% and 110.90%–3.57% higher than that of produced by other parameters. The highest *ACC* obtained of 75.82%, which is 10.96% to 0,37% maximum than different values of *lg*. Similarly, the *SEN* of 96.91% an acquired, which is 13.21% to 1.20% maximum than that achieved by different *lg* values.

Table 3 listed the evolution indicator obtained at the different *lag* values using NMBAC encoding scheme on human phosphorylated proteins dataset. It is illustrated in Table 3 that, when the *lag* value set 15, the *ACC* value reaches the highest of 72.20%, which is 5.67% to 0. 54% higher than that of *ACC* achieved by other parameters. Moreover, the highest values of *MCC* and AUC are 0.362 and 0.734, which are 28.82%–2.25% and 7.04%–0.13% maximum than the costs of *MCC* attained by other parameters. The maximum value of *SEN* is 90.80%, is acquired when the *lag* value set 35, and the highest *SPE* value is 45.14% when the *lag* value set 5. As we can observe from Fig. 2(c), when the *lag* value less than 15, it would lose some useful features, and when set *lag* value more significant than 15 may introduce noise. From the above observation, we can say that when the *lag* value set 15, the maximum peak of *MCC* and *AUC* values occurs, as can be seen from Fig. 2(c).

4.2. The impact of feature extraction methods on results

In developing computational models in the field of bioinformatics and computational biology, the use of feature extraction algorithms to abstract valuable information from protein sequences has become vital to the success or failure of the computational model [34]. In this study, three different feature extraction methods have been adopted to scramble the protein sequences into a mathematical expression. Generally, for a computational model, a single feature extraction method cannot detonate enough discriminative information from proteins sequence. Therefore, a different version of the hybrid feature and LFDA based optimal feature set was selected. This article, containing three distinct feature, i.e., (PsePSSM, PSSM-ACT, and NMBAC), different hybrid version feature, i.e., (PsePSSM + PSSM-ACT), all feature (PsePSSM + PSSM-ACT + NMBAC) and one optimum feature subset. To check the effectiveness and compare these five feature sets with the optimal feature set, SVM was employed as a classification engine to predict phosphorylated proteins with LOOCV test, and 10-fold cross-validation test. The prediction performance of each feature space is summarized in Table 4 obtained over both testing methods.

As illustrated in Table 4, some assumptions can be drawn. First, in case of individual features scheme over the jackknife test, the PSSM-ACT performed significantly better than two other single feature scheme, i.e., PsePSSM and NMBAC. The usage of SVM as a classification algorithm, the PSSM-ACT feature scheme succeeded a prediction performance with *MCC* of 0.476, an *ACC* of 76.27%, and *AUC* of 0.789 which is higher than the performance of PsePSSM and NMBAC characteristic features scheme respectively. Revisiting Table 4, it is confirmed that over the 10-fold cross-validations test again the PSSM-ACT outperformed single feature scheme. This phenomenon indicates that the PSSM-ACT feature has an enormous contribution to the prediction of the phosphorylated protein.

Secondly, the fusing of different features can further boost the prediction performance of the model. From Table 4, we found that the fused individual feature (PsePSSM + PSSM-ACT) outperformed the single features set, i.e., PsePSSM, PSSM-ACT, and NMBAC, using SVM over both LOOCV and 10-fold cross-validation test methods. On the other hands, NMBAC feature gives worse performance than the two other single features, i.e., PsePSSM and PSSM-ACT. As showed in Table 4, the NMBAC feature vector attained *MCC* of 0.355, an *ACC* of 71.92%, and an *AUC* of 0.727 using SVM as the as classifier over the jackknife test. However, when the NMBAC feature is combining the PsePSSM + PSSM-ACT feature, we found that the prediction performance significantly enhanced. As listed in Table 4, that all features (PsePSSM + PSSM-ACT + NMBAC) over both LOOCV test and 10-fold test using SVM classifier accomplished the superior performance with *MCC* of 0.527, an *ACC* of 78.93% and an *AUC* of 0.856, which 10.71%, 3.48%, and 8.49%, were better than the PsePSSM + PSSM-ACT feature respectively.

Thirdly, as we discuss in section 4.3 that the feature selection has a more significant impact, after comparing the performance of different feature selection methods, LFDA has outperformed others. Again, by revisiting Table 4, we can see that when we fed the LFDA optimized feature set to SVM classifier over the LOOCV test; it achieved *ACC* of 80.68%, *MCC* of 0.581 and *AUC* 0.867 which were higher than all feature space. In addition, equally metric results have been provided over the 10-fold cross-validation test. The ROC curve of the five feature sets and the optimal feature set (selected through LFDA) using jackknife test plotted in Fig. 3 (A). It is illustrated from Fig. 3(A) that ROC curve obtained by

Table 1
Prediction results by selecting different values of λ using the SVM classifier. Bold values indicate the best values.

PsePSSM	10-fold cross validation test (%)									
	Different values of λ									
	1	2	3	4	5	6	7	8	9	10
ACC	71.75	73.50	73.74	73.22	73.90	73.67	73.62	73.79	**74.69**	74.58
SEN	84.19	87.28	**87.99**	85.34	85.78	86.57	84.82	85.87	86.57	87.19
SPE	49.69	49.06	48.43	51.72	52.82	50.78	**54.70**	52.35	53.61	52.19
MCC	0.362	0.398	0.403	0.396	0.412	0.405	0.409	0.409	**0.430**	0.429
AUC	0.755	0.767	0.769	0.756	0.768	0.768	0.765	0.762	**0.770**	0.760

Table 2
Prediction results by selecting different values of *lg* using SVM classifier. Bold values indicate the best values.

PSSM-ACT	10-fold cross-validation test (%)									
	Different values of *lg*									
	1	2	3	4	5	6	7	8	9	10
ACC	67.51	68.64	**75.82**	75.54	74.29	74.35	71.75	71.47	72.15	71.86
SEN	92.93	90.72	**96.91**	94.43	95.76	88.16	87.19	86.04	85.60	87.72
SPE	22.41	29.47	38.40	42.20	36.21	**49.84**	44.36	45.61	48.28	43.73
MCC	0.222	0.260	**0.464**	0.448	0.421	0.418	0.338	0.349	0.368	0.355
AUC	0.680	0.708	**0.791**	0.778	0.784	0.773	0.743	0.744	0.743	0.740

Table 3
Prediction results by selecting different values of lag using SVM classifier. Bold values indicate the best values.

NMBAC	10-fold cross-validation test (%)									
	Different values of *lag*									
	5	10	15	20	25	30	35	40	45	50
ACC	70.23	71.58	**72.20**	71.81	71.75	70.11	68.32	69.21	68.98	69.32
SEN	84.36	88.69	88.69	87.54	88.87	86.48	**90.80**	86.04	84.45	90.02
SPE	45.14	**42.95**	42.95	43.89	41.38	41.07	32.13	39.34	41.54	32.60
MCC	0.321	0.348	**0.362**	0.354	0.350	0.312	0.281	0.289	0.288	0.281
AUC	0.715	0.734	**0.735**	0.709	0.720	0.698	0.684	0.686	0.692	0.702

Table 4
The performance of different features using SVM classifier over both 10-fold cross-validation and jackknife test. Bold values indicate the best values.

Features sets	Jackknife test					10-fold cross-validation test				
	ACC	SEN	SPE	MCC	AUC	ACC	SEN	SPE	MCC	AUC
PsePSSM	74.63	87.19	52.35	0.427	0.774	74.69	86.57	53.61	0.430	0.770
PSSM-ACT	76.27	**97.00**	39.5	0.476	0.789	75.82	96.41	38.40	0.464	0.791
NMBAC	71.92	88.52	42.48	0.355	0.727	72.2	88.69	42.95	0.362	0.735
PsePSSM + PSSM-ACT	77.06	88.34	57.05	0.485	0.804	76.55	88.25	55.8	0.473	0.794
All features[1]	78.93	91.43	56.74	0.527	0.856	77.68	**91.17**	53.76	0.498	0.849
Optimal features[2]	**80.68**	84.63	**73.67**	**0.581**	**0.867**	**79.43**	88.34	**63.64**	**0.544**	**0.861**

[1] All features = PsePSSM + PSSM-ACT + NMBAC.
[2] Optimal features = (PsePSSM + PSSM-ACT + NMBAC) +LFDA.

the optimal feature has the most substantial *AUC* value of 0.867, which are 1.28%, 7.83%, 12.66%, 10.51%, and 19.25% greater than that of the all feature, PsePSSM + PSSM-ACT, PsePSSM, PSSM-AC, and NMBAC methods, respectively. Based on these above observations, we employed the optimal feature set (PsePSSM + PSSM-ACT + NMBAC) +LFDA as the final feature vector with SVM classifier to construct the proposed HPhosPPred model for human phosphorylated proteins.

4.3. Analysis of different feature selection algorithms

In this portion of the paper, we will explain and compare the performance of four feature selection approaches; Adaptive Kernel Principal Component Analysis (AKPCA) [74], PCA (Principal Component Analysis) [75], Laplacian Eigenmaps [44] and LFDA to reduce dimensions and enhance the prediction performances of the model. Dissimilar prediction results were obtained by these feature reduction methods under different dimension sets of the phosphorylated proteins dataset, as shown in Table 5. We assess the discriminative power of feature subset (selected by different feature reduction methods) over the jackknife test and 10–fold cross-validation test techniques, and feature subset that achieves the highest *ACC*, *SEN*, *SPE*, *MCC*, and *AUC* values will be known as is the optimum feature subset. We employed SVM with RBF kernel as a learning machine to predict the optimal subset, selected by PCA, AKPCA, Laplacian, and LFDA. Different features subset selected by different feature selection methods over a 10-fold cross-validation test, detail of this subset can be seen in (supplementary file Table S2 and Figure S1). The list of best feature set indexes is listed in supplementary file Table S3 please see supplementary file.

As refer to Table 5, selecting different feature selection methods and dimension have a substantial influence on the performances of the model concerning the five assessment metrics, i.e., *SEN*, *SPE*, *MCC*, *ACC,* and *AUC*, over both LOOCV test and 10-fold cross-validation test. When the PCA and Laplacian Eigenmaps feature reductions methods were used to select an optimal features dimension, the feature size is 110-D, and the

Table 5
The performance comparison of prediction results of four feature selection methods over both 10-fold cross-validation and jackknife test. Bold values indicate the best values.

Feature selection	No. of Dimension of a feature vector	Jackknife test					10-fold cross-validation test				
		ACC	*SEN*	*SPE*	*MCC*	*AUC*	*ACC*	*SEN*	*SPE*	*MCC*	*AUC*
AKPCA	90-D	77.96	83.23	68.56	0.532	0.826	77.97	88.60	59.09	0.506	0.822
Laplacian	110-D	76.89	83.65	64.99	0.509	0.799	76.89	83.65	**64.99**	0.509	0.797
PCA	110-D	79.38	90.46	59.72	0.538	0.855	78.64	**90.46**	57.68	0.521	0.844
LFDA	70-D	**80.68**	**84.63**	**73.67**	**0.581**	**0.867**	**79.43**	88.34	63.64	**0.544**	**0.861**

highest *MCC* of 0.538 and 0.509, *ACC* of 79.38% and 76.89%, and *AUC* of 0.855 and 0.799 respectively. The AKPCA feature reduction method selected 90-D optimal feature and obtained the *MCC* of 0.532 *ACC* of 77.96% and *AUC* of 0.826. Furthermore, when the LFDA feature reduction scheme is applied to select the optimal feature, it selects 70-top feature from all features subspace with highest *MCC* of 0.581 of and *ACC* of 80.67%, an *AUC* of 0.867. Analogous results can also be achieved using a 10-fold cross-validation test, as showed in Table 5. Comparing the prediction performance obtained by different feature selection approaches with different dimensions size of feature, it can found that when the feature selection scheme LFDA is employed, it selects 70-dimension size of top feature set from original feature, and the prediction *ACC* was, 3.47%, 5.29%, and 1.62%, higher than AKPCA, Laplacian and PCA feature selection methods respectively. We conclude that the LFDA scheme has more obvious advantages over other feature reduction method, because the optimal feature subsets selected by LFDA, effectively minimize the time complexity, covering overfitting issue and improve the prediction performance of the model.

However, here a question arises; why the feature optimization improvest model prediction performance? To answer this question, the evaluation of the spatial distribution between all features and the optimal features was completed. To analyze the visual impact of reduced features, we use a graphical tool t-Distributed Stochastic Neighbor Embedding (t-SNE visualization) [76]. t-SNE program inevitable compress reduced feature vector to a two-dimensional space. The t-SNE display of all features and optimal features in the two-dimensional area drawn in Fig. 4(A) and (B) respectively, which shows that sample of phosphorylated proteins (label 1) and non-phosphorylated proteins (label −1) in all feature space are mixed-up and cannot be easily separated. However, after feature selection (LFDA), in optimal feature space, maximum samples of the phosphorylated proteins and non-phosphorylated proteins are scattered in two comparatively clear clusters. It proved that feature optimization could pick up the optimal features of the correct phosphorylated proteins and non-phosphorylated proteins.

4.4. The impact of kernel function on results

In the SVM classifier, the choice of kernel function has an indispensable influence on the prediction results as each kernel function utilized the distinct procedure for the transformation of data. After the selection of optimal feature set using LFDA, we examine the influence of different kernel function on the prediction performance, as shown in Table 6. These four functions kernel are linear, polynomial, RBF, and a sigmoid.

As illustrated in Table 6, the prediction effect of RBF and polynomial kernels are better than the linear and sigmoid kernel functions. RBF and polynomial obtained the highest prediction accuracies of 80.68% and 79.32%, respectively. Linear and sigmoid kernel function received ACC of 78.93% and 68.24%, respectively, which are 2.20% and 18.21% inferior to those achieved by RBF kernel. Furthermore, we compare the RBF kernel with other kernels on the ROC curve, as drawn in Fig. 3(B). It can be perceived from Fig. 3(B) that the area under the ROC curve acquired by the RBF kernel has the most substantial *AUC* value of 0.867, which are 3.91%, 20.92%, and 3.95% higher than that of all feature, linear, sigmoid and polynomial kernel respectively. It concludes that RBF is the better option to be used as a kernel function with our proposed model.

4.5. Comparison of SVM and other classification algorithms

In this section, the predictive performance of five-classification algorithms fixed-length, including RF, DT, LDA, and KNN with SVM using the optimal feature subset was compared. The SVM uses RBF function (c = 256, g = 256); in the RF classifier, 100 number of the decision tree is selected; the Euclidean distance and k neighbors value are set seven in KNN algorithm, LDA and DT algorithm used by default parameters. For comparison, we replace SVM in our model with these classification algorithms and re-construct the predictive models, which were then assessed over the LOOCV test and 10-fold cross-validation test on the same dataset. The details predictive performances of these classification algorithms can be found in Table 7.

As can be seen from Table 7, amongst the compared classification algorithms, different classifiers algorithm produces different predictive results. Using the SVM classifier for prediction of human phosphorylated proteins, the prediction rate of SVM is, *ACC* of 80.67%, *SEN* of 88.63%, *SPE* of 73.67%, *MCC* of 0.581, and *AUC* 0.867. The predicted accuracy of RF is 79.38%, DT is 77.51%, LDA is 76.50%, and KNN is 77.18%, which is 3.17%, 5.45%, 4.52%, and 4.07% respectively less than that of SVM algorithm. Besides, we draw the ROC curve of all five classifiers in Fig. 3(C), the *AUC* value of the SVM classifier is 0.867, which is 8.64%, 12.01%, 10.72%, and 19.42% greater than obtained by DT, KNN, LDA, and RF classifier respectively. After comparing the performances of the other four classifiers with SVM classifier, it can be summarized that the prediction performance of the SVM based model was significantly better than the different four classification algorithms in term of *SEN, SPE MCC, ACC,* and *AUC* over both jackknife test and 10-fold cross-validation test.

4.6. Comparison with a state-of-the-art method

To further evaluate the predictive performance of the proposed

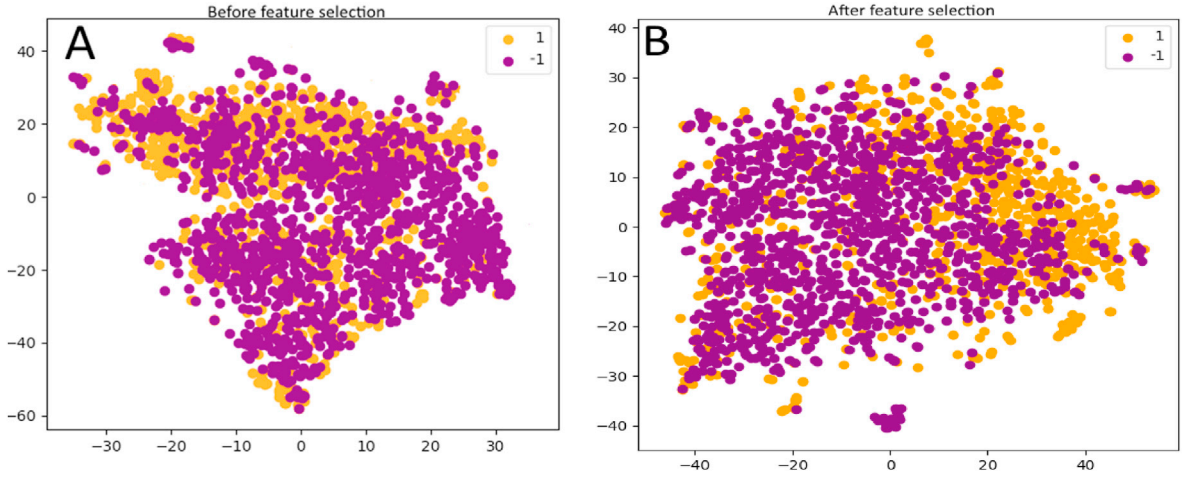


Fig. 4. t-SNE visualization in a two-dimensional feature space. (A) The original feature space (before feature selection), and (B) the optimum feature space (after feature selection).

Table 6
Prediction results of different kernel functions after feature reduction technique LFDA.

Functions types	Jackknife test					10-fold cross-validation test				
	ACC	*SEN*	*SPE*	*MCC*	*AUC*	*ACC*	*SEN*	*SPE*	*MCC*	*AUC*
Linear	78.93	**90.90**	57.69	0.527	0.843	78.14	87.28	61.91	0.513	0.841
Sigmoid	68.24	71.82	62.00	0.338	0.717	62.66	70.76	48.28	0.190	0.614
Polynomial	79.32	87.37	65.05	0.541	0.834	78.81	84.01	**69.59**	0.538	0.832
RBF	**80.68**	84.63	**73.67**	**0.581**	**0.867**	**79.43**	88.34	63.64	**0.544**	**0.861**

Bold values indicate the best values.

Table 7
The performance comparison of prediction results of five ML-algorithms algorithms over both 10-fold cross-validation and jackknife test. Bold values indicate the best values.

Classifiers	Jackknife test					10-fold cross-validation test				
	ACC	*SEN*	*SPE*	*MCC*	*AUC*	*ACC*	*SEN*	*SPE*	*MCC*	*AUC*
DT	77.51	82.86	68.03	0.510	0.798	77.06	81.45	**69.28**	0.504	0.788
KNN	77.18	**96.11**	43.57	0.493	0.774	76.72	**95.58**	43.26	0.481	0.795
LDA	76.50	86.22	59.25	0.475	0.783	75.20	85.60	56.74	0.445	0.546
RF	78.19	86.22	63.95	0.517	0.726	79.38	86.31	67.70	0.545	0.659
SVM	**80.68**	84.63	**73.67**	**0.581**	**0.867**	**79.43**	88.34	63.64	**0.544**	**0.861**

Table 8
A comparison of the proposed predictor with the existing method on the same dataset using both the jackknife test and 10-fold cross-validation test. Bold values indicate the best values.

Prediction model	Jackknife test				10-fold cross-validation test			
	ACC	*SEN*	*SPE*	*MCC*	*ACC*	*SEN*	*SPE*	*MCC*
iPhosPseEvo [6]	71.79	71.16	70.06	0.436	–	–	–	–
SKSOM [29]	78.75	78.45	**79.26**	0.561	–	–	–	–
HPhosPPred (Proposed)	**80.68**	**84.63**	**73.67**	**0.581**	**79.43**	**88.34**	**63.64**	**0.544**

predictor HPhosPPred, we compared the performance of our predictor with a recently published worked. To the best of our knowledge, in the literature, there is only two sequence-based predictor iPhosPseEvo and Ensembled SKSOM was developed so far as to predict human phosphorylated proteins for the current dataset. The values of the four metrics *(ACC, SEN, SPE, MCC)* achieved by predictor proposed in this study for the prediction of human phosphorylated proteins listed in Table 8. Table 8 also includes the corresponding metrics attained by iPhosPseEvo [6] and Ensembled SKSOM [29], and it should mention here that the performance of both predictors are noted directly from the previous prediction results. As shown in Table 8, that our proposed predictor HPhosPPred achieved the highest *ACC, SEN, SPE,* and *MCC* values of 80.67%, 84.63%, 73.67%, and 0.581, respectively. It indicates that our proposed predictor significantly outperforms the existing predictors on the same dataset. The experimental outcomes demonstrated that our model could be a useful bioinformatics tool for the prediction of human phosphorylated proteins.

Here, it is worthy to pinpoint what is the novelty of our work as compared to previous studies. To answer this question, firstly, in the proposed research, three nominal feature extraction methods (PsePSSM, PSSM-ACT, and NMBAC) have been used to better represent phosphorylated protein sequences by tuning different correlation coefficient parameters. Secondly, after comparing various feature reduction techniques performance, LFDA was implemented to efficiently eliminate the noise features, overcome the overfitting problem, and enhancing the generalization efficiency of the model. Thirdly, we compare the performance of four kernel functions. Altogether, the result demonstrates that HPhosPPred is a powerful prediction model, which can have produced not only superior or competitive results but also a robust predictive performance for human phosphorylated protein prediction.

5. Conclusion

Accurate prediction of human phosphorylated proteins is quite conducive to promote the knowledge about the underlying PTM regulatory mechanisms and disease treatment. In this paper, a new predictor, named PhosPPred, has been established for accurate identification of human phosphorylated proteins. We utilized three highly discriminative feature representation strategies, i.e., PsePSSM, PSSM-ACT, and NMBAC, based on evolutionary profile and physicochemical information from protein sequences. To remove the irrelevant information and enhance the prediction efficiency of the human phosphorylated proteins prediction model, LFDA was applied as an effective optimal feature selection technique on serially combined feature space. Lastly, the optimized feature set was fed into an SVM with the best kernel function RBF for training the proposed model. Thus, our developed HPhosPPred predictor, attained high prediction *ACC* of 80.67%, the *SEN* of 84.63%, the *SPE* of 73.67% and *MCC* of 0.581 using LOOCV test, similarly results over 10-fold cross-validation test were accomplished, i.e., *ACC* of 79.43%, *SEN* of 88.34% *SPE* of 63.64% and *MCC* of 0.544. The *ACC* and *MCC* of our method are 2.45% and 3.56% higher using Jackknife test than the recently published method for identification of human phosphorylated proteins in the literature so far. The experimental results reveal that the developed model produced quite encouraging success rates than the existing system due to several reasons including practical feature engineering, suitable feature reduction method and robust classifier with different kernels. The comparative analysis shows that the proposed predictor HPhosPPred has achieved better performance. It is anticipated that the proposed predictor will provide support in designing the new computational research for the prediction of human phosphorylated proteins in particular and new pharmaceutical drug development in general.

In recent publications [1,71,77] a freely accessible and user-friendly web-servers have been launched that obviously improve the performance of a newly developed method. It is the future direction to establish user-friendly and publicly accessible web-servers and databases. Thus, we will try our best to establish such webserver for maximizing the convenience of most experimental scientists. The source code and dataset are publicly available at https://github.com/saeed344/HPhosPPred.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemolab.2020.104066.

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